

Cosmic.jl

A cosmology and Einstein–Boltzmann engine in pure Julia

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github.com/aburousan/Cosmic.jl

Equations follow D. Baumann, *Cosmology* (CUP, 2021)

What this talk is about

I made a code called **Cosmic.jl**. It is written fully in **Julia**. It does the same job as CLASS and CAMB, but in one language.

In this talk I will show:

- What a Boltzmann code actually solves (little bit of physics and maths).
- Why I wrote one more code, when CLASS and CAMB are already very good.
- Some features, with small code examples.
- How close my results are to CLASS and CAMB.
- What is still missing. I will say this part honestly.

One thing I want to say early. CLASS and CAMB are the **references** here. My code tries to reproduce them, not the other way. Where my numbers are different, I assume my code is wrong until I can prove otherwise.

The physics: two problems, one after another

1. Smooth universe

How fast is the universe expanding? This comes from the Friedmann equation. If we write every component as a density, then

$$H^2(a) = H_0^2 \sum_s \frac{\rho_s(a)}{\rho_{c,0}}$$

Each component s (photons, baryons, dark matter, neutrinos, dark energy) only needs to know its own $\rho_s(a)$.

2. Small ripples

On top of the smooth part there are small perturbations. Photons are described by a distribution function, and it follows the Boltzmann equation

$$\frac{df}{d\eta} = C[f]$$

where $C[f]$ is Thomson scattering off free electrons. (Baumann, App. B.1)

Everything a Boltzmann code prints — the CMB spectrum, the matter power spectrum — comes from these two problems, plus one projection integral.

From ripples to what we observe

We expand the photon temperature field in multipoles $\Theta_\ell(k, \eta)$. After decoupling the photons just free-stream. Then the observed angular power spectrum is one integral over k :

$$C_\ell = 4\pi \int \frac{dk}{k} \underbrace{P_{\mathcal{R}}(k)}_{\text{transfer function}} |\Theta_\ell(k, \eta_0)|^2$$

↑
primordial spectrum from inflation (Baumann §8.3)

So there are exactly three pieces:

1. **Initial conditions** — what inflation gives us.
2. **Evolution** — the Boltzmann + Einstein system, until today.
3. **Projection** — turning a 3D k into a 2D angle ℓ (Baumann §7.3.1, App. B.2).

A Boltzmann code is basically a careful machine for these three steps.

Why write one more code?

CLASS (C) and CAMB (Fortran) are excellent and very well tested. I did not write Cosmic.jl to replace them. I wrote it for three reasons.

- 1. One language from top to bottom.** In CLASS and CAMB the fast part is C or Fortran, and the part you actually use is a Python wrapper. Julia is fast enough to be both. So the code you read is the code that runs.
- 2. Composition should be generic.** A cosmology here is just a **list of species**. To add a new component you write one function, its $\rho(a)$. You do not touch ten other files.
- 3. Compute, do not fit.** Many codes use fitting formulae for hard integrals. Where I can, I compute the actual integral, and keep the fit only as a cross-check.

The design, in one screen

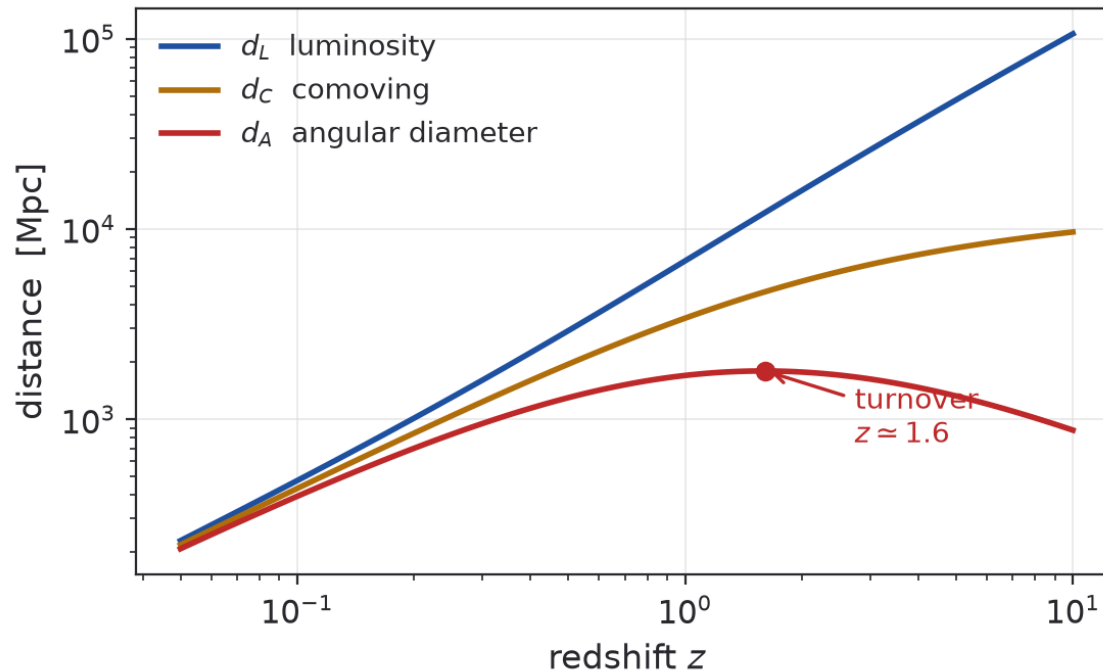
using Cosmic

```
c = cosmology()           # Planck 2018 LCDM
age(c)                    # 13.787 Gyr
luminosity_distance(c, 1.0) # Mpc

rec = recombination(c)    # thermal history
sp  = cmb_spectra(c, rec; lmax=1500, nk=20000)
mp  = matter_power_spectrum(c, rec; kmin=1e-3, kmax=10.0)
```

That is the whole interface for a standard run. `cosmology()` builds the species list, `recombination` solves the thermal history, and the last two do the perturbations and the projection.

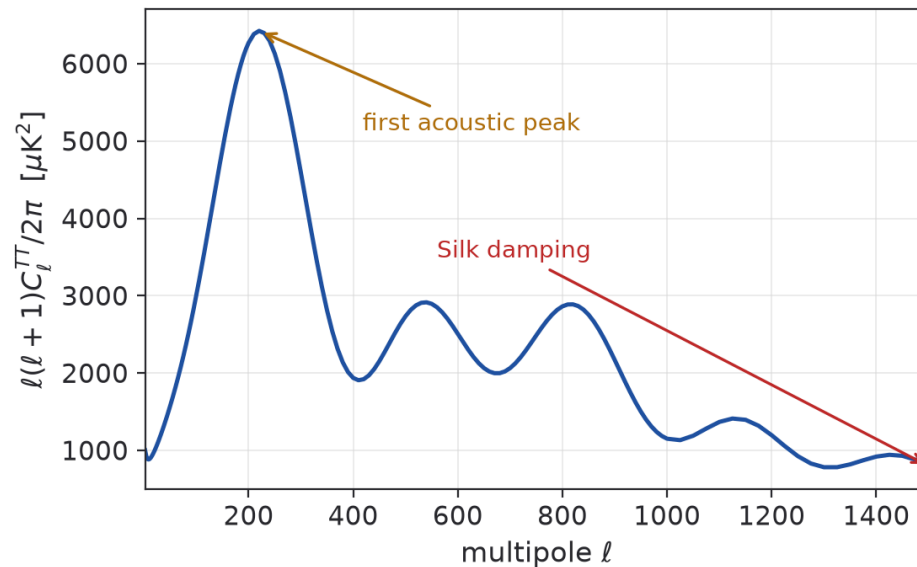
Distances: the easy check first



Luminosity distance d_L , angular diameter distance d_A , comoving distance d_C . Note d_A turns over near $z \approx 1.6$: far away objects start looking **bigger** again, because the universe was smaller when the light left.

Against CLASS these agree to about 10^{-9} .

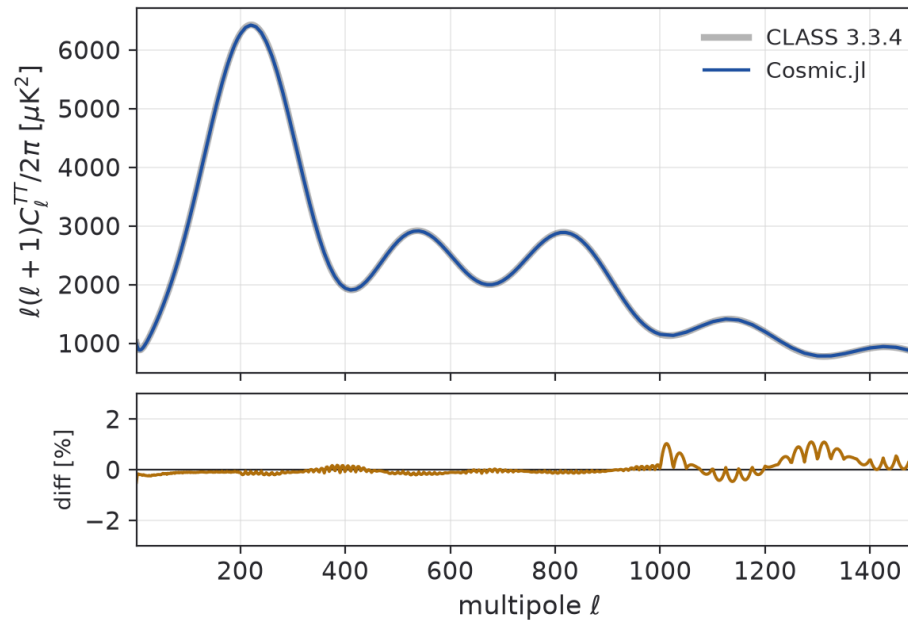
The CMB spectrum, and what the bumps mean



Before decoupling, photons and baryons are one fluid. Gravity pulls it in, radiation pressure pushes it out. This gives **sound waves** (Bau-mann §7.4).

- The **first peak** is the mode that had just enough time for one compression.
- Peak **positions** tell us the geometry.
- Peak **heights** tell us how much baryon and dark matter there is (§7.5.1–7.5.2).
- The fall at high ℓ is **Silk damping**: photons diffuse out of small overdensities (§7.4.4).

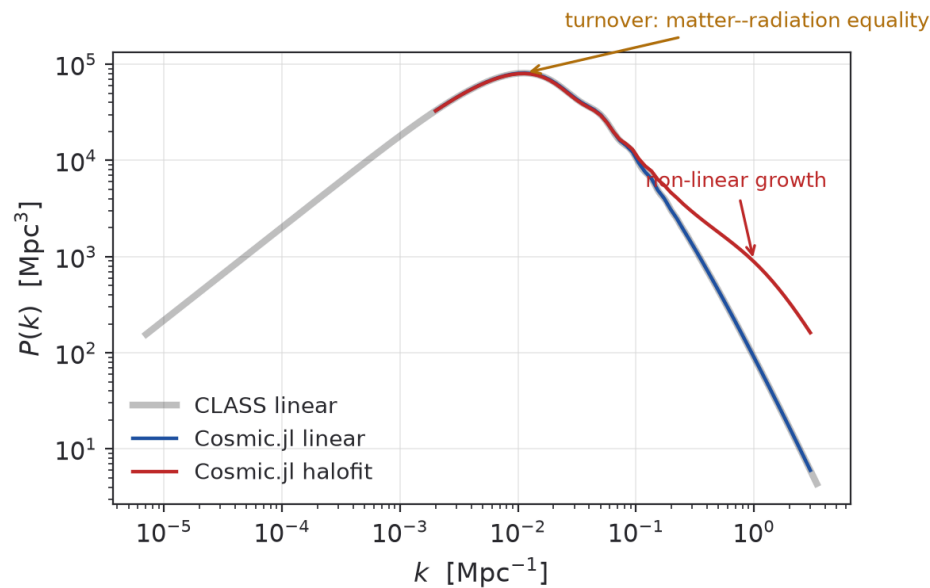
Cosmic.jl compared with CLASS



Same cosmological parameters on both sides, same Y_p , and no reionization on either side. Top: the two curves sit on top of each other. Bottom: the difference in percent. It stays under **0.5%** below $\ell \approx 1000$, and under 1.1% up to $\ell = 1500$.

First make it converge, then compare. My first try at this plot had 8% ringing near the first peak. It was not physics. The code itself had warned me that the k grid was too coarse ($\Delta k = 7.9 \times 10^{-5}$, but 1.2×10^{-5} was needed). After raising nk to 20000 the ringing was gone. So always converge both codes before you call a difference a bug.

The matter power spectrum



$P(k)$ tells us how clumpy matter is at scale k (Baumann §5.3.4).

- It rises, then turns over. The turnover is the scale that entered the horizon at matter-radiation equality.
- Modes entering earlier grow more slowly, because radiation stops them. So small scales are suppressed.
- The small wiggles are the same sound waves as in the CMB.
- At large k gravity is no longer linear, so we need a non-linear correction (halofit here).

Some features I like

BBN is solved, not looked up. CLASS reads Y_p from a table, CAMB uses a fitting formula. Here the 12-nuclide, 63-reaction network is integrated in the same background, so the helium that goes into recombination is the helium your own ω_b produced.

Spectral distortions from first principles. Exact Karzas–Latter Gaunt factor, exact double-Compton emissivity, and the exact Klein–Nishina $\otimes$ Maxwell–Jüttner kernel instead of the Kompaneets approximation.

Curvature done exactly. Hyperspherical Bessel functions are solved from their own recurrences at every ν . CLASS switches to rescaled flat Bessels above $\nu = 4000$; here that switch never happens.

Early-universe gravitational waves. Scalar-induced GWs and primordial black hole abundances, including non-Gaussianity (f_{NL} , g_{NL}).

CLASS, CAMB and Cosmic.jl side by side

	CAMB	CLASS	Cosmic.jl
First release	2000	2011	in development
Language	Fortran 90	C	Julia
Interface	Python wrapper	classy wrapper	native
Recombination	RECFAST, HyRec, CosmoRec	RECFAST, HyRec	RECFAST-class, HyRec-2020
BBN	fitting formula	interpolated table	solved network
Curvature	exact	flat approx. above $\nu = 4000$	exact at all ν
Non-linear	halofit, HMcode	halofit, HMcode	halofit, HMcode, EFT
Distortions	—	fit based	first principles
Maturity	very high	very high	young

How close are the numbers?

Cross-checks at **matched settings**, against live CLASS 3.3.4 and CAMB runs:

Quantity	Reference	Agreement
Distances	CLASS	10^{-9}
$P(k)$, σ_8	CAMB	0.04%, 0.007%
C_ℓ^{TT} unlensed	CLASS	$\approx 0.4\%$
lensed TT / EE	CLASS	0.3% / 0.4%
$C_\ell^{\varphi\varphi}$, $\ell \leq 1000$	CLASS	$\leq 0.9\%$
halofit $P_{\text{nl}}/P_{\text{lin}}$	CLASS	0.25%
EFT 1-loop	FAST-PT	$< 0.5\%$
Y_p	CLASS (PRIMAT)	$< 0.1\%$
Y_p , D/H, ^7Li	PRyMordial	-0.004% , $+0.03\%$, -1.1%
μ , y distortions	CLASS	0.02%
open / closed C_ℓ^{TT}	CLASS / CAMB	$\leq 0.6\%$ / $\leq 0.4\%$

What is missing, and what is approximate

I think a code is easier to trust if it says its own weak points.

Not implemented yet: vector modes; number-count and shear spectra; late-time y -distortion; scalar-field and interacting dark energy perturbations; a full second-order (bispectrum) solver.

A known approximation. For tensor and vector modes I use the “optimal” Boltzmann hierarchy, which re-expands them on scalar normal modes. Pitrou, Pereira & Lesgourgues (PRD **102**, 023511) proved that the separability this depends on **fails when the space is curved**. For scalars the error is tiny, but for **tensor polarisation** it is about $50 |\Omega_K| \%$.

So curved-space tensor EE and BB in my code are not exact. I keep this written down instead of hiding it.

Summary

- Cosmic.jl solves the full chain in one language: background $\rightarrow$ BBN $\rightarrow$ recombination $\rightarrow$ perturbations $\rightarrow$ CMB and $P(k)$.
- It agrees with CLASS and CAMB at the few $\times 10^{-3}$ level on distances, C_ℓ , $P(k)$, BBN abundances and spectral distortions.
- What makes it different: species-based design, exact curved geometry, a real BBN network, and distortion physics computed instead of fitted.
- It is still young. CLASS and CAMB stay the references, and the open approximations are written down, not hidden.

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